# Overview of R and its use in research

# Learning objective

To give a general background of R software, its features and use in research

# Let’s Talk R!

- R is a free open source programming language for statistical computing and data visualization.
- Created by **R**oss Ihaka and **R**obert Gentleman in 1992 (University of Auckland, New Zealand).
- Yes, R stems from the first letter of their first names, inspired by another programming language called S.
- R is useful when conducting statistical analysis and visualizations in different fields.
- Where there is a lot of data to **wrangle**, R is the best option.

## Features of R

- **Comprehensive statistical tools** for data analysis (descriptive statistics, regression, time-series, spatial analysis).
- **Powerful visualization capabilities** through packages like `ggplot2`, enabling clear and impactful data communication.
- **Integration with databases and other languages** such as Python and SQL for seamless data workflows.
- **Reproducible research** supported by tools like **R Markdown** and **Quarto**.

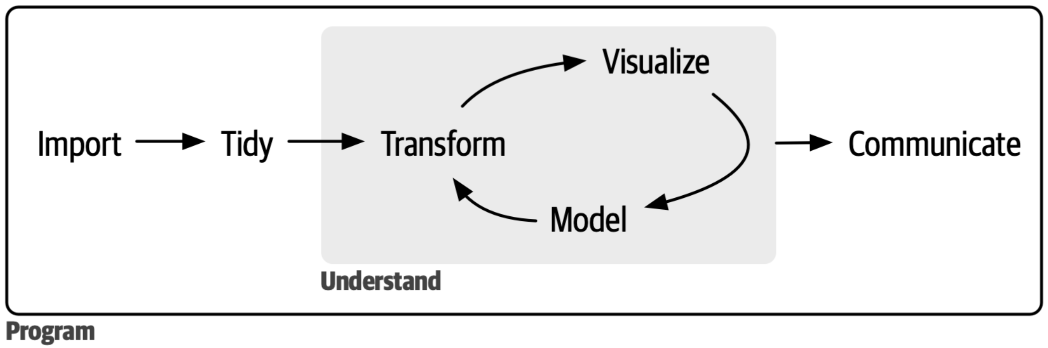
# Relevance in Health Surveillance

- **Data Management:** Efficiently handles large health datasets from multiple sources (e.g., DHIS2, registries, laboratory systems).
- **Epidemiological Analysis:** Enables modeling of disease trends, outbreak detection, and estimation of incidence or prevalence.
- **Geospatial Analysis:** Supports mapping of disease distribution and identification of hotspots using packages like `sf` and `tmap`.
- **Time-Series and Forecasting:** Monitors epidemic curves and predicts future trends (e.g., COVID-19, malaria).
- **Visualization and Reporting:** Produces automated dashboards, reports, and visual summaries for timely decision-making.
- **Reproducibility:** Ensures analyses can be replicated and updated as new data become available.

# What you will learn

Our aim in this training is to help you master the key R tools for performing analysis efficiently and reproducibly - and enjoy the learning process along the way. Our framework for the stages of a typical data analys using R can be represented as shown in the following figure.

In this model, you start with **data import** and **tidying**. Next, you understand your data with an iterative cycle of **transforming**, **visualizing**, and **modeling**. You finish the process by **communicating** your results to other humans. Surrounding all these tools is **programming**.



# The R Ecosystem

The **R ecosystem** refers to the entire network of *tools, packages, communities, and resources* that support working with the **R programming language** — especially for data analysis, visualization, and statistical computing.

## Base R

Base R comes with a set of recommended packages that are automatically installed when you install R

```
14 base packages
 - Create R objects
 - Summaries
 - Maths functions
 - Statistics
 - Graphics
 - Datasets
```

## Contributed packages

The main package repository is **CRAN (Comprehensive R Archive Network)**, but others exist like **Bioconductor** (for bioinformatics) and **GitHub** (for experimental or custom tools).

### CRAN

- Official R repository
- https://cran.r-project.org

- Over 22,900 packages

## Bioconductor

- Bioinformatics packages
- https://www.bioconductor.org
- ~2340 packages

## GitHub

- Packages in development
- GitHub-only packages

What can R do?

- Data import
- Data management and wrangling
- Exploratory data analysis
- Statistical modelling
- Advanced statistics

- Machine learning
- Data visualisation
- Reports, articles
- Dashboards, web apps
- Integrates well with other languages
- Packages: share your code and use others